Part B – Formulas & Short Theory

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1. Bernoulli Distribution

Definition

Bernoulli Distribution is a discrete probability distribution with only two possible outcomes:

- Success = 1
- Failure = 0

It is used for single trial experiments.

Formula

$$P(X = x) = p^x(1 - p)^{1-x}$$

Where:

- p= probability of success
- $1 - p$= probability of failure

Example

Transaction successful or failed.

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2. Binomial Distribution

Definition

Binomial Distribution calculates the probability of getting a fixed number of successes in multiple independent trials.

Formula

$$P(X = x) = \binom{n}{x} p^x(1 - p)^{n-x}$$

Where:

- n= total number of trials
- x= number of successes
- p= probability of success

Example

Number of successful transactions in one week.

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3. Poisson Distribution

Definition

Poisson Distribution models the number of events occurring within a fixed interval of time.

Formula

$$P(X = x) = \frac{e^{-\lambda}\lambda^{x}}{x!}$$

Where:

- λ= average number of events
- x= number of occurrences

Example

Number of transactions per day.

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4. Log-Normal Distribution

Definition

A Log-Normal Distribution occurs when the logarithm of a variable follows a normal distribution.

It is positively skewed and used for financial data.

Formula

$$f(x) = \frac{1}{x\sigma\sqrt{2\pi}} e^{-\frac{(\ln x - \mu)^2}{2\sigma^2}}$$

Example

Transaction amounts and income data.

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5. Power Law Distribution

Definition

Power Law Distribution describes situations where small values occur frequently and large values occur rarely.

Formula

$$P(x) \propto x^{-a}$$

Where:

- a= scaling parameter

Example

Large transaction amounts occurring less frequently.

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6. Q-Q Plot

Definition

Q-Q Plot (Quantile-Quantile Plot) is used to compare dataset distribution with theoretical normal distribution.

Interpretation

- Points near straight line → normal distribution
- Points away from line → non-normal distribution

Use

Used for testing normality of transaction amounts.

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7. Box-Cox Transform

Definition

Box-Cox Transformation stabilizes variance and improves normality of data.

Formula

$$Y(\lambda) = \frac{Y^{\lambda} - 1}{\lambda}$$

Where:

- λ= transformation parameter

Use

Used to reduce skewness in transaction data.

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8. Z-Score

Definition

Z-Score measures how far a value is from the mean in terms of standard deviation.

Formula

$$Z = \frac{X - \mu}{\sigma}$$

Where:

- X= observed value

- μ= mean
- σ= standard deviation

Interpretation

- Positive Z-score → above mean
- Negative Z-score → below mean

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9. Probability Density Function (PDF)

Definition

PDF shows the density of probability for continuous random variables.

Formula

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Use

Used to analyze probability distribution of transaction amounts.

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10. Cumulative Distribution Function (CDF)

Definition

CDF gives cumulative probability up to a specific value.

Formula

$$F(x) = P(X \leq x)$$

Interpretation

Shows probability that a variable takes value less than or equal to x.

Use

Used for cumulative transaction probability analysis.